

Bhawna Sinha

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Summary

Strong foundation in RF/mmWave and THz frequency with experience in patch arrays, EM modeling, high-frequency measurement, and system-level integration with hands-on spectrum analyzer, signal generator and THz systems.

Education

Master of Science	expected Jun 2026
Electrical and Computer Engineering, University of California, Davis	GPA: 3.6/4
Bachelor of Technology	Aug 2018 - Jun 2022
Electrical and Electronics Engineering, Cochin University of Science and Technology, India	GPA: 9.05/10

Skills

- **Languages:** Python, C++, C, MATLAB, SQL
- **Domain:** RF/mmWave, Electromagnetics, Optics, Photonics, Analog circuit, Antenna design, Meta- material, PCB design
- **Software:** ANSYS HFSS, Lumerical, COMSOL, Altium, LTSpice, Keysight ADS, AutoCAD, LabVIEW
- **RF Equipment:** Spectrum Analyzers, Network Analyzers, Signal Generators, Pump and Probe
- **Wireless Communication Protocols:** 5G nR, GSM, CDMA, LTE, Bluetooth, GPS

Experience

- Graduate Researcher**, Applied Micro/Nano Electromagnetics Research Laboratory, UC Davis
RF, EM, MEMS, Antenna design Sep 2023 - Present
- Designed a patch array antenna and resonator, focusing on simulation and performance optimization for microwave applications.
 - Performing modeled tunable RF behavior in 2D photonic crystals through material doping, voltage control, and simulation-driven analysis.
 - Working on synchronized THz and infrared (IR) signal generation using pump-probe techniques, aimed at enabling ultra-fast spectroscopy and advanced signal characterization.
 - Achieved control and manipulation of electromagnetic waves, using theoretical modeling and computational analysis.
- Lead Engineer**, TATA Power, Mumbai, India — Testing, SCADA Jun 2022 - Jul 2023
- Developed an SMS-connected Breaker Trip Coil Condition Monitoring System.
 - As a team responsible for the installation of 145 kV Hyosung Make Gas Insulated Switchgear.
 - Led the automation of maintenance and testing checklists for high-voltage handling equipment, ensuring uninterrupted 24/7 power supply across voltage levels from 22kV to 220kV, even under adverse conditions.
 - Successfully designed and deployed an automated robotic solar panel cleaner, improving panel efficiency by up to 75%.
- Teaching Assistant**, UC Davis Apr 2024 – Present
- Led lab sessions on filter and amplifier design, supporting students with systematic troubleshooting, circuit-level debugging, and measurement-based validation.

Academic Projects

- Rectangular Waveguide Bandpass Filter** Sep 2025
- Designed an X-band rectangular-waveguide (WG-15) bandpass filter using an EM-driven filter design approach in both reflection and transmission configurations.
 - Extracted equivalent circuit parameters and developed a SPICE-based model to represent the filter's resonant and coupling behavior.
 - Validated the design through SPICE-HFSS co-simulation, demonstrating strong agreement in S-parameters, cutoff characteristics, and guided-wave propagation of the dominant TE_{10} mode.
 - Extended the design by introducing bias-dependent tuning to shift the resonant frequency, analyzing its impact on waveguide dispersion and filter response
- Patch Array Antenna Design for mmWave Applications** Dec 2025
- Designed a 7 GHz patch array antenna targeted for 5G wireless networks, achieving improved gain and beam-steering capability through array optimization.
 - Performed impedance matching and bandwidth tuning, validating results with S-parameter and far-field analysis.
 - Conducted parametric sweeps to evaluate array performance under fabrication tolerances and environmental variations.
- Patch Array Metasurface Resonator: Reflection characteristics at low-terahertz frequencies** ongoing
- Designed a high-Q microwave resonator operating at 500 GHz, optimized for strong field confinement and enhanced sensitivity.
 - Developed equivalent circuit models in LTSpice and validated against full-wave simulations in HFSS.
 - Demonstrated workflow integration between circuit- and EM-level modeling, applicable to antenna design.